

Yaxiong Liu

✉ yxliu269@cqut.edu.cn, yaxiongliu@amss.ac.cn

🌐 <https://yaxiongliu23.github.io/>

Citizenship: China, Sex: Male, Date of birth: February 28, 1995.

Research Interests

Kähler geometry, Algebraic geometry, especially K-stability of polarized varieties, Several complex variables and complex geometry, especially singular metric on vector bundles.

Education

Tsinghua University – Beijing, China

September 2020 – June 2023

PhD in Mathematics

Supervisor: Professor Akito Futaki.

Institute of Mathematics, Academy of Mathematics and

Systems Science, Chinese Academy of Sciences – Beijing, China

September 2017 – June 2020

M.S. in Mathematics

Supervisors: Professor Weidong Ruan, Professor Jiayu Li.

Lanzhou University – Lanzhou, China

September 2013 – June 2017

B.S. in Mathematics and Applied Mathematics

Employment

Chongqing University of Technology, CN

September 2026 –Present

Specially Appointed Associate Professor

University of Maryland – College Park, US

August 2023 – July 2026

Novikov Postdoc in Mathematics

Mentor: Professor Tamás Darvas.

Honors and Scholarships

2022

- National Scholarship for Graduate Student (China)
- Excellent Postgraduate Award (Yau Mathematical Sciences Center in Tsinghua University)

2021

- Excellent Postgraduate Award (Yau Mathematical Sciences Center in Tsinghua University)
The scholarship is supported by Professor Yau, the so-called Ruolin graduate Award, named by his mother.
- Comprehensive Scholarship (Tsinghua University)

Publications and Preprints

- With J. Chu, *Alternative proof of the gradient estimate for complex Hessian equations*, arXiv:2609.33091.
- With J. Chu, N. McCleerey and W. Zhang, *Some variational problems for the complex Monge–Ampère operator*, arXiv:2604.13421.
- With J. Han, *$\mathbb{G}$ -uniform weighted K-stability for models on klt varieties*, arXiv:2506.18039.
- With Z. Liu and Hui Yang, *An asymptotic formula for the Segre classes*, **J. Geom. Anal.** 35 (2025), no 3, 101.
- With J. Han, *On the existence of weighted-cscK metrics*, **Adv. Math.** 463 (2025), Paper No. 110125.
- With J. Chu and N. McCleerey, *The eigenvalue problem for the complex hessian operator on m -pseudoconvex manifolds*, **J. Funct. Anal.** 290(3) (2026), Paper No. 111256.
- With Zhuo Liu, Hui Yang, and Xiangyu Zhou, *A Le Potier-type isomorphism twisted with multiplier submodule sheaves*, **J. Math. Pures Appl.** 202 (2025), Page No. 103760.

- Y. Liu, *Openness of uniformly valuative stability on the Kähler cone of projective manifolds*, **Math. Z.** 303 (2023), no 2, 52.
- Y. Liu, *Uniform K-stability and conformally Kähler, Einstein-Maxwell geometry on toric manifolds*, **Tohoku Math. J. (2)** 74 (2022), no 1, 1-21.

Teaching and TA Experiences

Teaching

- Department of Mathematics, University of Maryland
 - Math-436: *Differential geometry of curves and surfaces I*, Fall, 2025
 - Math-401: *Applications of Linear Algebra*, Fall, 2025
 - Math-401: *Applications of Linear Algebra*, Spring, 2025
 - Math-121: *Elementary Calculus II*, Fall, 2024
 - Math-140H: *Calculus I*, Fall, 2024
 - Math-401: *Applications of Linear Algebra*, Spring, 2024
 - Math-436: *Differential geometry of curves and surfaces I*, Fall, 2023

TA

- School of Mathematical Sciences, Tsinghua University
 - *Complex Geometry*, Spring, 2023.
 - *Topics in Complex Geometry*, Fall, 2022.
 - *Differential geometry III–Complex Geometry*, Spring, 2022.
 - *Differential geometry III–Complex Geometry*, Spring, 2021.
- School of Mathematical Sciences, University of Chinese Academy of Sciences
 - *Differential geometry*, Fall, 2018.
 - *Calculus III*, Fall, 2017.

Academic Visits

- July 26 – September 13 2026, Westlake University, visiting Prof. Jiyuan Han.
- July 2025, Mathematical Science Research Center, Chongqing University of Technology.
- July 2024, Westlake University, visiting Prof. Jiyuan Han.
- May 2024, Rutgers University, visiting Prof. Jian Song for one week.
- June 2023, Westlake University, visiting Prof. Jiyuan Han for one week.
- February 2017–June 2017, Enhanced Program for Graduate Study, Beijing International Center For Mathematical Research.

Invited Talks

- 9 2026, The variational approach on the Dirichlet problem for the complex Monge–Ampère equation, at Westlake University.
- 7 2026, The variational approach on the Dirichlet problem for the complex Monge–Ampère equation, Informal Several Complex Variables seminar, Chongqing University of Technology.
- 3 2026, Existence of weighted cscK metrics, geometry seminar at George Mason University.
- 7 2025, Existence of weighted cscK metrics, at Chongqing University of Technology.
- 7 2025, Existence of weighted cscK metrics, Workshop on Geometric Analysis 2025 at Qingdao
- 6 2024, The eigenvalue problem for the complex hessian operator on m-pseudoconvex manifolds, ITS seminar, Westlake University
- 6 2024, The existence of weighted cscK metrics and G-coercivity of weighted Mabuchi functional, Beijing Normal University
- 6 2024, The eigenvalue problem for the complex hessian operator on m-pseudoconvex manifolds, Geometry & Topology seminar at Chongqing University of Technology
- 6 2024, The eigenvalue problem for the complex hessian operator on m-pseudoconvex manifolds, 2024 Geometry & Topology conference at Qingtian
- 5 2024, A Le Potier-Type Isomorphism Twisted with Multiplier Submodule Sheaves, Geometric Analysis Seminar, Peking

- 3 2024, The eigenvalue problem of complex Hessian operators, Differential Geometry Seminar, Tsinghua
- 10 2023, A Le Potier-type Isomorphism with Multiplier Submodule Sheaves, Geometry and Geometric Analysis seminar in Purdue
- 10 2023, The eigenvalue problem of complex Hessian operators, IGA seminar, UMD
- 6 2023, Recent progress on the valuative stability of polarized varieties, ITS seminar, Westlake University
- 1 2023, Uniformly valuative stability of polarized varieties and applications, Complex Analysis and Geometry Seminar, Rutgers
- 12 2022, Uniformly valuative stability of polarized varieties and applications, IGA seminar, Maryland University
- 11 2022, Uniformly valuative stability of polarized varieties and applications, Tsinghua University
- 12 2020, Uniform K-stability and cKEM geometry on toric manifolds, Chinese Academy of Sciences.
- 12 2020, Geodesic rays and stability in the cscK problem, Beijing Geometric Analysis Day in Xiangshan.
- 10 2020, Uniform K-stability and cKEM geometry on toric manifolds, Peking University.